

Networks: supply and demand

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Solution of the practice quiz

1. Remember that the divergence of a flow vector $x \in \mathbb{R}^n$ at node i is defined as the total quantity of flow that leaves the node minus the total quantity of flow that enters the node:

$$\operatorname{div}(x)_i = \sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki}.$$

We obtain:

- $\operatorname{div}(x)_1 = (2.3 + 4) - 0 = 6.3,$
 - $\operatorname{div}(x)_2 = (-1 + 3) - (-2.1 + 2.3) = 1.8,$
 - $\operatorname{div}(x)_3 = (4.5) - (4 - 1) = 1.5,$
 - $\operatorname{div}(x)_4 = (-5 - 2.1) - (3 - 5) = -5.1,$
 - $\operatorname{div}(x)_5 = (-5) - (-5 + 4.5) = -4.5,$
 - $\operatorname{div}(x)_6 = (2.5) - (3) = -0.5,$
 - $\operatorname{div}(x)_7 = (3) - (2.5) = 0.5.$
2. Nodes 1, 2, 3 and 7 have a positive divergence. They are supply nodes. Nodes 4, 5 and 6 have a negative divergence. They are demand nodes. There is no transit node.
 3. The sum of the divergences in the network is *always* zero. It can be checked that

$$6.3 + 1.8 + 1.5 - 5.1 - 4.5 - 0.5 + 0.5 = 0.$$

4. Here, the cut $\Gamma = (\mathcal{M}, \mathcal{N} \setminus \mathcal{M})$ is defined as $\mathcal{M} = \{1, 2, 3\}$ and $\mathcal{N} \setminus \mathcal{M} = \{4, 5, 6, 7\}$.
The set of forward arcs is

$$\Gamma^{\rightarrow} = \{(2, 4), (3, 5)\},$$

and the set of backward arcs is

$$\Gamma^{\leftarrow} = \{(4, 2)\}.$$

The flow through the cut Γ is defined as

$$X(\Gamma) = \sum_{(i,j) \in \Gamma^{\rightarrow}} x_{ij} - \sum_{(i,j) \in \Gamma^{\leftarrow}} x_{ij}.$$

Then we obtain

$$X(\Gamma) = (4.5 + 3) - (-2.1) = 9.6.$$

The sum of the divergences of the nodes belonging to $\mathcal{M}$ is

$$\sum_{i \in \mathcal{M}} \text{div}(x)_i = 6.3 + 1.8 + 1.5 = 9.6.$$

Thus the flow through Γ is equal to the sum of the divergences of the nodes belonging to $\mathcal{M}$. This is always the case.

The intuition is the following. Consider an arc (i, j) such that both i and j are in $\mathcal{M}$. When calculating the sum of the divergences in $\mathcal{M}$, the flow of this arc will be involved once with a positive sign for the divergence of j , and once with a negative sign for the divergence of i . Therefore, its contribution to the sum is zero. It means that only arcs (i, j) such that $i \in \mathcal{M}$ and $j \in \mathcal{N} \setminus \mathcal{M}$ will contribute to the sum. But these are exactly the arcs belonging to the cut. Therefore,

$$\sum_{i \in \mathcal{M}} \text{div}(x)_i = \sum_{(i,j) \in \Gamma^{\rightarrow}} x_{ij} - \sum_{(i,j) \in \Gamma^{\leftarrow}} x_{ij} = X(\Gamma).$$